10.7 Gravity Inside a Shell

WORKSHEET

The question

STEP III 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q9

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Consider two small, diametrically opposite pieces of the ring. Draw the forces they exert on the particle. Which components cancel, and which add?

STAGE 2 Slicing the shell into rings

Ask yourself Put the centre of the shell at O and the particle on the axis at distance b from O. Slice the shell into thin rings perpendicular to the axis, labelled by the angle ϕ that a point of the ring makes with the axis at O. What are the radius, the position and the mass of the ring?

STAGE 3 The right variable

Ask yourself The denominator involves $s^2 = a^2 - 2ab\cos\phi + b^2$. What does s represent geometrically? What is $\mathrm{d}s$ in terms of $\mathrm{d}\phi$?

STAGE 4 **It vanishes**

Pause and try What changes if the particle is *outside* the shell, with $b > a$? The limits of s become $b - a$ and $b + a$. Redo the last line and show that the force is $\frac{GM}{b^2}$ towards the centre, as if all the mass were concentrated there.

What's really going on

Newton's cone argument

Newton's own proof (*Principia*, Book I, Proposition 70) uses no integration at all. Draw a narrow double cone with its vertex at the particle, cutting the shell in two small patches (Figure 10.7.1). The areas of the patches are proportional to the squares of their distances from the particle, r_1^2 and r_2^2 (the patches make equal angles with the cone, since a chord meets a circle at equal angles at its two ends). So their masses are proportional to r_1^2 and r_2^2, and their pulls, proportional to mass divided by distance squared, are *equal* and opposite. Every double cone through the particle cancels, so the total force is zero. The inverse-square law is exactly what makes the cancellation work: with any other power of distance, the shell would push or pull.

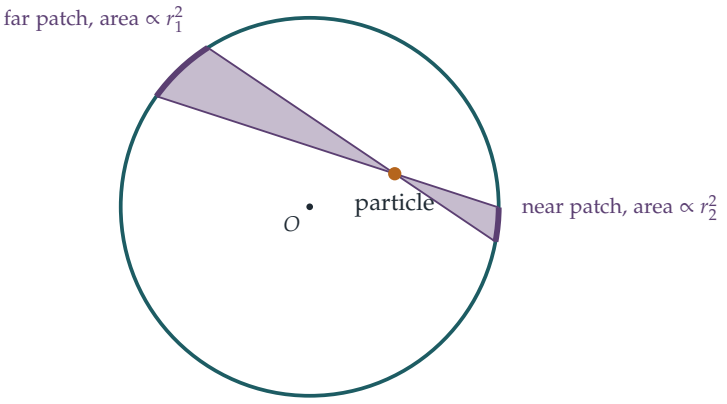


Figure 10.7.1 Newton's argument. A thin double cone from the particle cuts the shell in two patches. The far patch is bigger, by the ratio of the squared distances, but its pull is weaker by exactly the same ratio. The two pulls cancel, and so does every other pair: inside the shell, gravity vanishes.

Collapsing clouds

STEP III 1987 Q14 uses the theorem the other way round. Outside a uniform sphere, the pull on a particle at the surface is GMm/r^2, as if the whole mass were at the centre. Now take a uniform spherical cloud of dust, at rest, of radius R and mass M, and let it collapse under its own gravity. A particle on the surface feels only the mass inside it, which stays M as everything falls inwards, so its inward speed $v = -\dot{r}$ satisfies

$$v\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{GM}{r^2}, \qquad \text{so} \qquad v^2 = 2GM\Big(\frac{1}{r} - \frac{1}{R}\Big).$$

Substituting $r = R\cos^2\alpha$ (one of the exercises) gives the time to reach radius r as $\sqrt{\frac{R^3}{2GM}}\big(\alpha + \frac{1}{2}\sin 2\alpha\big)$. The whole cloud collapses to a point when $\alpha = \frac{\pi}{2}$, after the *free-fall time*

$$t_{\mathrm{ff}} = \frac{\pi}{2}\sqrt{\frac{R^3}{2GM}} = \sqrt{\frac{3\pi}{32G\rho}},$$

where ρ is the density. Remarkably, every shell of the cloud reaches the centre at the same moment, since the time depends only on the density (Figure 10.7.2). For a typical interstellar cloud, with about 10^{-17} kg per cubic metre, t_{ff} is around 700 000 years: this sets the timescale on which stars form.

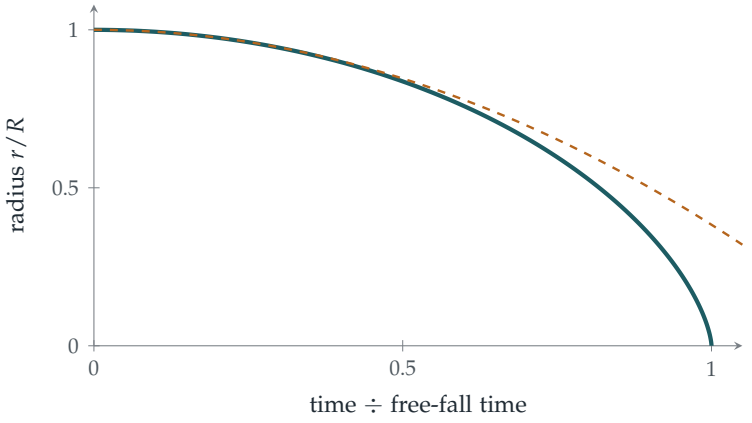


Figure 10.7.2 Radius of a collapsing uniform dust cloud against time (teal). It starts slowly, like a dropped stone (the dashed amber parabola is the constant-acceleration approximation $r = R - \frac{1}{2}\frac{GM}{R^2}t^2$), then accelerates sharply and reaches zero at $t = t_{\mathrm{ff}}$. The curve is half of a cycloid.

Newton's delay and Gauss's law

Newton had the inverse-square law by the 1660s, but did not publish his theory of gravity until 1687. One reason, according to some historians, was that he could not justify treating the Earth as a point mass until he had proved the shell theorem; once he had, the rest of the *Principia* followed. Today the theorem is a short consequence of *Gauss's law*, which says that the total gravitational field flowing out through any

closed surface is proportional to the mass inside. Take a sphere inside the shell, with the same centre. It encloses no mass, so the total flow through it is zero. That alone does not make the field zero (a uniform field has zero total flow through any closed surface, flowing in on one side and out on the other), but here symmetry helps: the field must point straight towards or away from the centre, with the same strength everywhere on the sphere, so zero total flow means zero field. The same law governs electric fields, which is why there is no electric field inside a charged metal shell, and why your car protects you from lightning.

Try this next

1. **Outside the shell.** Complete the Pause box: show that outside the shell the force is $\frac{GM}{b^2}$. Deduce that a solid uniform sphere of mass M pulls an outside particle as if its mass were at the centre.

2. **The collapse time.** Try STEP III 1987, Question 14 (step.maths.org/questions/87-s3-q14). It finds how long a cloud of dust takes to collapse under its own gravity; a trigonometric substitution does the integral.

3. **A tunnel through the Earth.** Inside a uniform solid sphere, show that the force at distance b from the centre is $\frac{GMb}{a^3}$, proportional to b. Deduce that a ball dropped down a tunnel through the centre of the Earth oscillates with period $2\pi\sqrt{a^3/(GM)}$, about 84 minutes, the same as the period of a satellite skimming the surface.

4. **Another force law.** If gravity followed an inverse-cube law, would the force inside a shell still be zero? (Repeat Stage 3 with $\frac{1}{s^3}$ in place of $\frac{1}{s^2}$.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapters on gravitation, including the shell theorem.
- **Subrahmanyan Chandrasekhar, *Newton's Principia for the Common Reader*** (Oxford University Press). Newton's proofs, including Propositions 70 and 71, explained by a Nobel laureate.
- **Andrew Pontzen, *The Universe in a Box*** (Riverhead). How gravity, collapse and simulation explain the structure of the universe.

The full guided solution is in Section 10.7:
https://beyondstep.keithmcnulty.org/ch10/shell-theorem.html